Sunil Verma

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Summary

Data Engineer with **5+ years** of experience in designing and optimizing scalable data architectures using Azure and Databricks. Experienced in leading end-to-end pipeline development, enhancing performance, and delivering impactful data solutions. Focused on engineering excellence, continuous learning, and building robust, future-ready data systems that support advanced analytics and business innovation.

Skills

- **Data Engineering & Pipelines:** Azure Data Factory, Azure Databricks, Delta Lake, ADLS
- **Big Data Technologies:** Hive, Spark (PySpark), Spark SQL, Spark Streaming
- **Programming & Scripting:** SQL, Python
- **Cloud & Tools:** Microsoft Azure, Unity Catalog, GitHub, Kafka
- **Emerging Tech & Practices:** Generative AI, Agentic AI, Agile Methodology, Performance Optimization

Work Experience

Tata Digital – *Lead Data Engineer*

Oct 2020 – Present | Bangalore

- Built an autonomous **AI-powered monitoring and auto-healing** system for Databricks workflows, enabling intelligent failure detection and automated remediation with minimal downtime.
- Led the design and orchestration of a **config-driven ingestion framework** supporting **incremental and near real-time loads** from **PostgreSQL, MongoDB, Cosmos DB, and SFTP**, ingesting data into **Delta Lake** on **Azure Databricks**, with built-in **data quality validations** to improve processing performance and ingestion success rate by **~95%** and ensure high reliability in production.
- Led the design of an **AI-driven cost optimization framework** that combines **Databricks System Tables** with **Open AI powered analysis** to assess **cluster utilization efficiency**, recommend optimal resource sizing, and support **ongoing reduction in cloud compute spend**.
- Designed an **enterprise reconciliation framework** automating reconciliation across **multiple source systems**, enabling **40+ automated reconciliations**, cutting **manual effort and errors by up to 90%**, and improving overall settlement efficiency.
- Developed a **scalable Feature Mart** comprising **5K+ curated features** across **multiple Tata brands**, standardizing feature definitions and pipelines, reducing **data scientists' feature engineering time by ~70%**, and enabling automated **model execution and refresh pipelines**.
- Led performance optimization of **PySpark pipelines** by converting full data processing to **Delta-based incremental loads** and **liquid clustering**, significantly improving query performance, while enforcing **Unity Catalog–based governance**, RBAC, and a **central audit layer** for enterprise compliance.
- Led the design of a **secure, PII-compliant data enrichment framework** supporting data exchange with **multiple organizations**, implementing **double encryption and audit controls**, automating enrichment for **10M+ records**, cutting **manual effort by ~70%**, and improving overall processing time by **~80%**.
- Manage and mentor a team of **2 junior** data engineers, driving technical excellence through comprehensive code reviews, architecture design, and implementation of coding standards across all project phases

Education

National Institute of Technology Tiruchirappalli
Master of Computer Applications | **CGPA**: 7.42 | 2017 – 2020

Key Achievements

Recognized for driving innovation and delivering high-impact data engineering solutions. **Trailblazer Award – Exceeding** Everest *(Mar 2025)*

Certifications

- **Azure Databricks & Spark Core for Data Engineer - May 2021**
- **Azure Data Factory for Data Engineer - Mar 2022**